

Failure (an endless loop, and a real pause/resume) — Solution

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This exercise has 2 separate halves — the endless loop, and the checkpoint/interrupt/resume cycle. Each version below covers both. Read all three depths — they're not "wrong, less wrong, right," they're 3 real, valid ways to solve the same problem, with real tradeoffs between them.

Basic Version

Approach 1 — the direct way

Part 1 — the endless loop, stopped by a low recursion limit:

```
from typing import TypedDict
from langgraph.graph import StateGraph, START, END
from langgraph.errors import GraphRecursionError

class LoopState(TypedDict):
    count: int

def node_a(state):
    return {"count": state["count"] + 1}

def node_b(state):
    return {"count": state["count"] + 1}

builder = StateGraph(LoopState)
builder.add_node("node_a", node_a)
builder.add_node("node_b", node_b)
builder.add_edge(START, "node_a")
builder.add_edge("node_a", "node_b")
builder.add_edge("node_b", "node_a")
```

```
graph = builder.compile()

try:
    graph.invoke({"count": 0}, config={"recursion_limit": 5})
except GraphRecursionError as e:
    print(f"Stopped on purpose after hitting the recursion limit: {e}")
```

Expected output:

```
Stopped on purpose after hitting the recursion limit: Recursion limit of 5 reached
```

(The exact wording of the message can vary slightly by LangGraph version — the important, reliable part is the exception type, `GraphRecursionError`, and that it fires instead of the process hanging forever.)

Part 2 — checkpoint, interrupt, resume:

```
from typing import TypedDict
from langgraph.graph import StateGraph, START, END
from langgraph.checkpoint.memory import MemorySaver
from langgraph.types import interrupt, Command

class ApprovalState(TypedDict):
    task: str
    approved: bool

def do_task(state):
    decision = interrupt({"question": f"Approve running: {state['task']}?"})
    return {"approved": decision}

builder = StateGraph(ApprovalState)
builder.add_node("do_task", do_task)
builder.add_edge(START, "do_task")
builder.add_edge("do_task", END)

checkpointer = MemorySaver()
graph = builder.compile(checkpointer=checkpointer)

config = {"configurable": {"thread_id": "run-1"}}

first_result = graph.invoke({"task": "send an email", "approved": False}, config=config)
print("after first invoke:", first_result)
```

```
second_result = graph.invoke(Command(resume=True), config=config)
print("after resume:", second_result)
```

Expected output:

```
after first invoke: {'task': 'send an email', 'approved': False, '__interrupt__':
after resume: {'task': 'send an email', 'approved': True}
```

The first `invoke()` runs `do_task`, which calls `interrupt(...)` and pauses right there — the graph never reaches `END` on that call, and the returned state carries the interrupt's payload instead of a finished result. The second `invoke()`, using `Command(resume=True)` and the **same** `thread_id`, picks `do_task` back up from exactly where `interrupt()` paused it — `decision` becomes `True`, the function finishes, and `approved` ends up `True` in the final state.

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Intermediate Version

Approach 1 — type hints, and reading the interrupt payload properly

Part 1:

```
from typing import TypedDict
from langgraph.graph import StateGraph, START, END
from langgraph.errors import GraphRecursionError

class LoopState(TypedDict):
    count: int

def node_a(state: LoopState) -> dict:
    return {"count": state["count"] + 1}

def node_b(state: LoopState) -> dict:
    return {"count": state["count"] + 1}

builder = StateGraph(LoopState)
builder.add_node("node_a", node_a)
builder.add_node("node_b", node_b)
builder.add_edge(START, "node_a")
builder.add_edge("node_a", "node_b")
builder.add_edge("node_b", "node_a")

graph = builder.compile()

try:
    final_state = graph.invoke({"count": 0}, config={"recursion_limit": 5})
    print(f"Somehow finished: {final_state}") # should never happen with no exit
except GraphRecursionError:
    print("Confirmed: an exit-less loop hits the recursion limit and fails loudly")
```

Expected output:

Confirmed: an exit-less loop hits the recursion limit and fails loudly, not silently

Part 2:

```
from typing import TypedDict
from langgraph.graph import StateGraph, START, END
from langgraph.checkpoint.memory import MemorySaver
from langgraph.types import interrupt, Command

class ApprovalState(TypedDict):
    task: str
    approved: bool

def do_task(state: ApprovalState) -> dict:
    decision = interrupt({"question": f"Approve running: {state['task']}?"})
    return {"approved": decision}

builder = StateGraph(ApprovalState)
builder.add_node("do_task", do_task)
builder.add_edge(START, "do_task")
builder.add_edge("do_task", END)

checkpointer = MemorySaver()
graph = builder.compile(checkpointer=checkpointer)
config = {"configurable": {"thread_id": "run-1"}}

paused_state = graph.invoke({"task": "send an email", "approved": False}, config=
pending_interrupts = paused_state.get("__interrupt__", [])
if pending_interrupts:
    question = pending_interrupts[0].value["question"]
    print(f"Graph paused, asking: {question}")
else:
    print("Graph did not pause -- something's wrong with the interrupt wiring")

resumed_state = graph.invoke(Command(resume=True), config=config)
print(f"Final state after resume: {resumed_state}")
```

Expected output:

Graph paused, asking: Approve running: send an email?

```
Final state after resume: {'task': 'send an email', 'approved': True}
```

Difference from Basic: Part 1 adds a check that the graph genuinely never finishes on its own (the `try` block's success branch is unreachable by design, which is itself worth confirming). Part 2 reads the actual interrupt payload out of `paused_state["__interrupt__"]` and prints the real question instead of just printing the whole raw state dict — this is what a real approval UI would do: show the human the *question*, not a debug dump.

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Advanced Version

Approach 1 — proving `MemorySaver` does NOT survive a real restart

This is 2 separate scripts, run one after the other, with the process exiting completely in between — not 2 function calls in the same script.

`pause_script.py` :

```
from typing import TypedDict
from langgraph.graph import StateGraph, START, END
from langgraph.checkpoint.memory import MemorySaver
from langgraph.types import interrupt

class ApprovalState(TypedDict):
    task: str
    approved: bool

def do_task(state: ApprovalState) -> dict:
    decision = interrupt({"question": f"Approve running: {state['task']}?"})
    return {"approved": decision}

builder = StateGraph(ApprovalState)
builder.add_node("do_task", do_task)
builder.add_edge(START, "do_task")
builder.add_edge("do_task", END)

checkpointner = MemorySaver()
graph = builder.compile(checkpointer=checkpointner)
config = {"configurable": {"thread_id": "run-1"}}

result = graph.invoke({"task": "send an email", "approved": False}, config=config)
print("Paused. Now exit this process entirely and run resume_script.py separately")
print(result)
```

`resume_script.py` (run as a brand-new `python resume_script.py`, after `pause_script.py` has already exited):

```

from typing import TypedDict
from langgraph.graph import StateGraph, START, END
from langgraph.checkpoint.memory import MemorySaver
from langgraph.types import interrupt, Command

class ApprovalState(TypedDict):
    task: str
    approved: bool

def do_task(state: ApprovalState) -> dict:
    decision = interrupt({"question": f"Approve running: {state['task']}?"})
    return {"approved": decision}

builder = StateGraph(ApprovalState)
builder.add_node("do_task", do_task)
builder.add_edge(START, "do_task")
builder.add_edge("do_task", END)

# a brand-new MemorySaver -- this process never saw the first invoke() at all
checkpointer = MemorySaver()
graph = builder.compile(checkpointer=checkpointer)
config = {"configurable": {"thread_id": "run-1"}}

resumed_state = graph.invoke(Command(resume=True), config=config)
print(resumed_state)

```

Expected output from `resume_script.py` :

```

Traceback (most recent call last):
...
langgraph.errors.EmptyInputError: ... (or an equivalent error / empty-state result)

```

This is the point, not a mistake to fix: `MemorySaver` really does keep its data only in the Python process that created it. A genuinely separate process has a genuinely empty checkpointer, so there's nothing to resume — proving hands-on that `MemorySaver` is a development/testing tool, not something that survives an actual restart.

Approach 2 — a checkpointer that actually persists, using SQLite

```

from typing import TypedDict
from langgraph.graph import StateGraph, START, END

```

```

from langgraph.checkpoint.sqlite import SqliteSaver
from langgraph.types import interrupt, Command

class ApprovalState(TypedDict):
    task: str
    approved: bool

def do_task(state: ApprovalState) -> dict:
    decision = interrupt({"question": f"Approve running: {state['task']}?"})
    return {"approved": decision}

builder = StateGraph(ApprovalState)
builder.add_node("do_task", do_task)
builder.add_edge(START, "do_task")
builder.add_edge("do_task", END)

config = {"configurable": {"thread_id": "run-1"}}

# pause_script.py
with SqliteSaver.from_conn_string("checkpoints.sqlite") as checkpointer:
    graph = builder.compile(checkpointer=checkpointer)
    graph.invoke({"task": "send an email", "approved": False}, config=config)
    print("Paused, saved to checkpoints.sqlite. Safe to exit now.")

# resume_script.py -- a genuinely separate process, run later
with SqliteSaver.from_conn_string("checkpoints.sqlite") as checkpointer:
    graph = builder.compile(checkpointer=checkpointer)
    resumed_state = graph.invoke(Command(resume=True), config=config)
    print(resumed_state)

```

Expected output (from the second, genuinely separate process, run any time after the first one exited):

```
{'task': 'send an email', 'approved': True}
```

Same graph, same interrupt, same `Command(resume=True)` call — the only change is swapping `MemorySaver()` for `SqliteSaver.from_conn_string("checkpoints.sqlite")`, which writes each checkpoint to an actual file on disk. That one substitution is the entire difference between "survives a second function call in the same script" and "survives the process exiting, the computer rebooting, or a completely different program resuming it hours later."

Difference from Intermediate, and between these 2 Advanced approaches: Intermediate proves pause/resume works within one running script, using `MemorySaver`. Approach 1 deliberately tries the same thing across 2 truly separate processes and shows it fails — the checkpointer's data was only ever in that first process's RAM. Approach 2 fixes the actual problem Approach 1 exposes, by swapping in `SqliteSaver`, a checkpointer backed by a real file — proving the exact same graph and interrupt code now genuinely survives a restart, with no other changes needed.

Which one should you actually use? `MemorySaver` for local development and this document's exercises — it's fast, needs no setup, and is exactly right for testing a graph's logic. The moment a pause needs to survive past the current process — a real human approval workflow, a server that might restart, anything actually going to production — swap to a real persisted checkpointer (`SqliteSaver` for something simple and local, `PostgresSaver` for a real deployed service). Approach 1's failure is worth actually watching once, the same way the endless loop in Part 1 is — so "why didn't my resume work" never becomes a confusing surprise in a real project.